

Project Plan

Due 10/19/2023

1. Group:

- a. Chris Davis Jaldi – U01099335
- b. Aditya Mallakula – U01093160

2. Project Dataset:

- a. **Link to dataset:** <https://www.kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link/data>

- b. **Target (if supervised):**

The target variable is the ‘Level’ of lung cancer – “High”, “Medium” or “low”.

- c. **Features:**

The 24 features are as follows:

Age, Gender, Air Pollution, Alcohol use, Dust Allergy, Occupational Hazards, Genetic Risk, chronic Lung Disease, Balanced Diet, Obesity, Smoking, Passive Smoker, Chest Pain, Coughing of Blood, Fatigue, Weight Loss, Shortness of Breath, Wheezing, Swallowing Difficulty, Clubbing of Fingernails, Frequent Cold, Dry Cough, Snoring

- d. **Brief description:**

The lung cancer patient dataset contains measurements on 24 features related to demographics, lifestyle factors, symptoms etc. for 1000 patients. The target variable (Level) indicates low, medium, or high cancer level. This can be used to build models to predict cancer level from the input features. The input features include age, gender, lifestyle factors, symptoms, etc.

3. Timeline:

Date	Student	Tasks	Description
09/25/2023	Chris/Aditya	Project Plan	Develop a comprehensive project plan (Primary project and dataset decision factors)
10/08/2023	Chris/Aditya	Exploratory Data Analysis	Analyze dataset characteristics through EDA for initial data processing plans.
10/11/2023	Chris/Aditya	Preprocessing	Handle missing data, outliers, and duplicates using ML techniques

10/20/2023	Aditya	Model Development	Build Supervised Models A (Simple ML techniques such as Logistic Regression, etc.)
	Chris		Build Supervised Models B (Complex ML techniques such as Neural Networks, etc.)
11/20/2023	Chris/Aditya	Model Evaluation and Interpretation	Evaluate model performance and create insightful visualizations
11/23/2023	Chris/Aditya	Testing and Validation	Conduct testing and validate the final model's performance
11/30/2023	Chris/Aditya	Documentation and Reporting	Summarize key findings and insights using valid performance metrics and create a presentation video.

Note: While the timeline may change, this will provide a rough idea of the tasks that may need more (or less time). Please leave room for the final project report writing. That will be the outcome/ evaluation for the final project points for this class.